

Lecture 1.4 – Intro to R and R Studio

Specific Learning Objectives:

1.1.1 – Understand how to use the command line.

1.1.2 – Understand how to use the help function of R.

1.1.3 – Understand the basic syntax of the R language.

1.1.4 – Execute inbuilt mathematical functions to perform calculations in R.

1.1.5 – Learn how to assign variables.

1.1.6 – Understand the basic syntax of functions in R.

1.1.7 – Open, edit, and save a script in RStudio's editor.

2.2.1 – Create reproducible scripts in R.

About *R*



- A high-level interpreted language designed primarily to run statistical tests and visualize data.
 - Based on the language S, which is not used anymore.
 - Free, open-source language (meaning anyone can access the base code all the way down to compilers!)
 - Base *R* code is written in C, C++, Fortran, and *R*.
 - Interpreted language: meaning you don't worry about compiling machine code, it happens automatically in the background.
 - *R* installations come with a simple command line and script editor.

About R Studio



- A Integrated Development Environment (IDE) for the *R* language.
- R Studio is basically a wrapper with some helpful features! It is not *R* itself, which has to be downloaded separately.
- Helpful features include:
 - Command-line prompt
 - Help viewer
 - Integrated plot editor
 - Jobs & testing environments
 - Memory management tools
 - File system navigator
 - Package Manager
 - Script Editor
 - Integrated RMarkdown & Knitr manager
 - Version Control panel

A Tour of R Studio

The image shows the RStudio interface with several annotations:

- Script Editing Window**: Points to the main editor window containing the R script `install_dotwalkr.R`.
- Environment variables**: Points to the Environment pane on the right, which shows the Global Environment.
- Current R Project**: Points to the Project: (None) dropdown in the top right corner.
- File system navigator**: Points to the Files pane on the right, which shows the file structure of the current project.
- Command-line console**: Points to the Console window at the bottom, which displays the R prompt and output.

The R script `install_dotwalkr.R` contains the following code:

```
1 #### Install script ####
2
3 #### Required Packages ####
4 message("Testing for required packages, installing if not found.")
5 message(" ")
6 packages <- c("testthat", "R.matlab", "data.table", "parallel", "doParallel", "foreach")
7
8 package.check <- lapply(
9   packages,
10   FUN <- function(x) {
11     if (!require(x, character.only = TRUE)) {
12       message(paste("Installing Package ", x, " now."))
13       install.packages(x, dependencies = TRUE, repos='http://cran.us.r-project.org')
14       library(x, character.only = TRUE)
15     } else {
16       message(paste("Package ", x, " found."))
17     }
18   }
19 )
20 message("Done with required packages!")
21 message(" ")
22
```

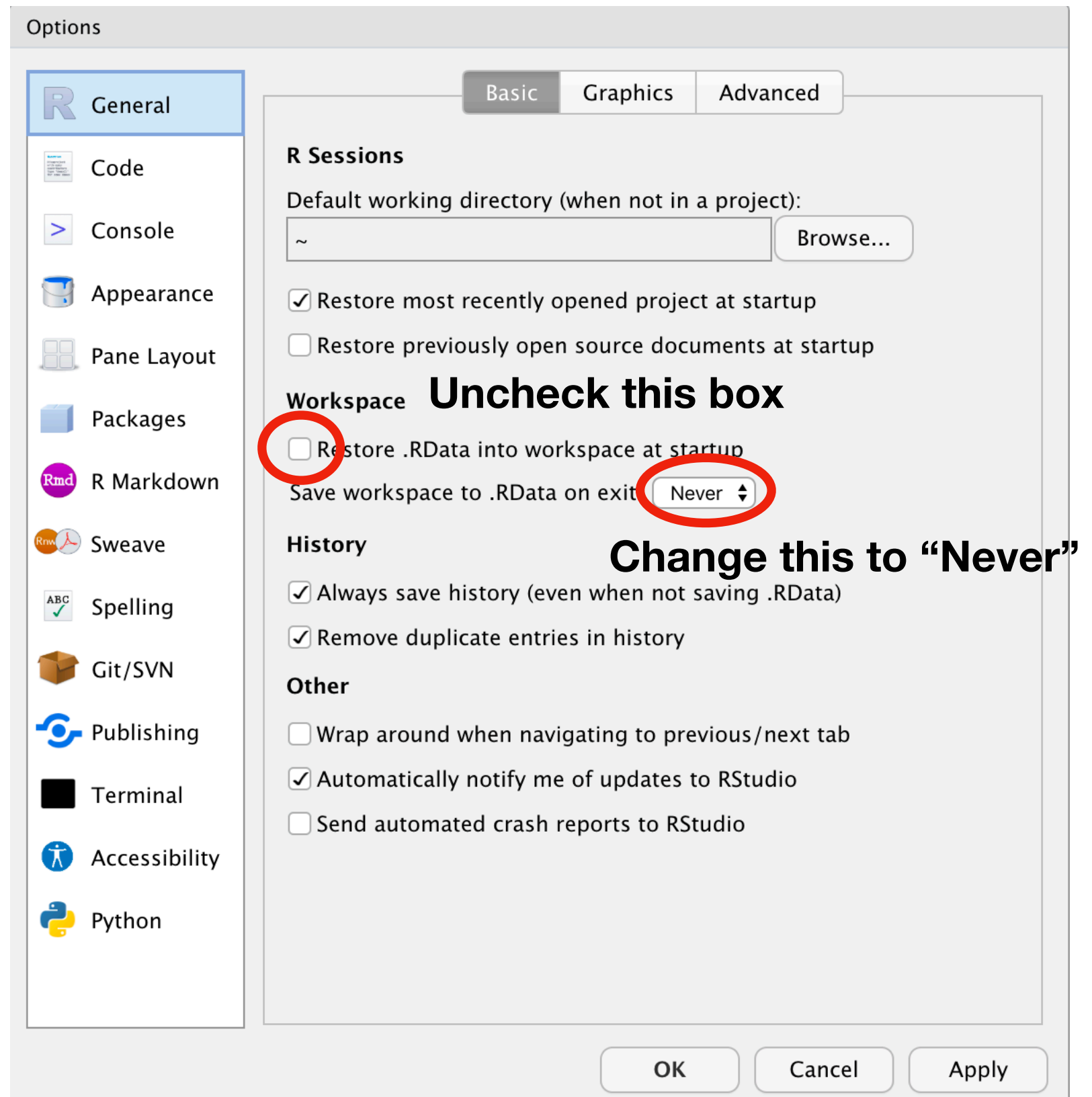
The Environment pane shows the Global Environment, which is empty.

The Files pane shows the file structure of the current project, including files like `.gitignore`, `.Rhistory`, `data`, `doc`, `dotwalkR.Rproj`, `install_dotwalkr.R`, `README.md`, `results`, `Rplot.pdf`, `Rplot01.pdf`, `src`, and `tests`.

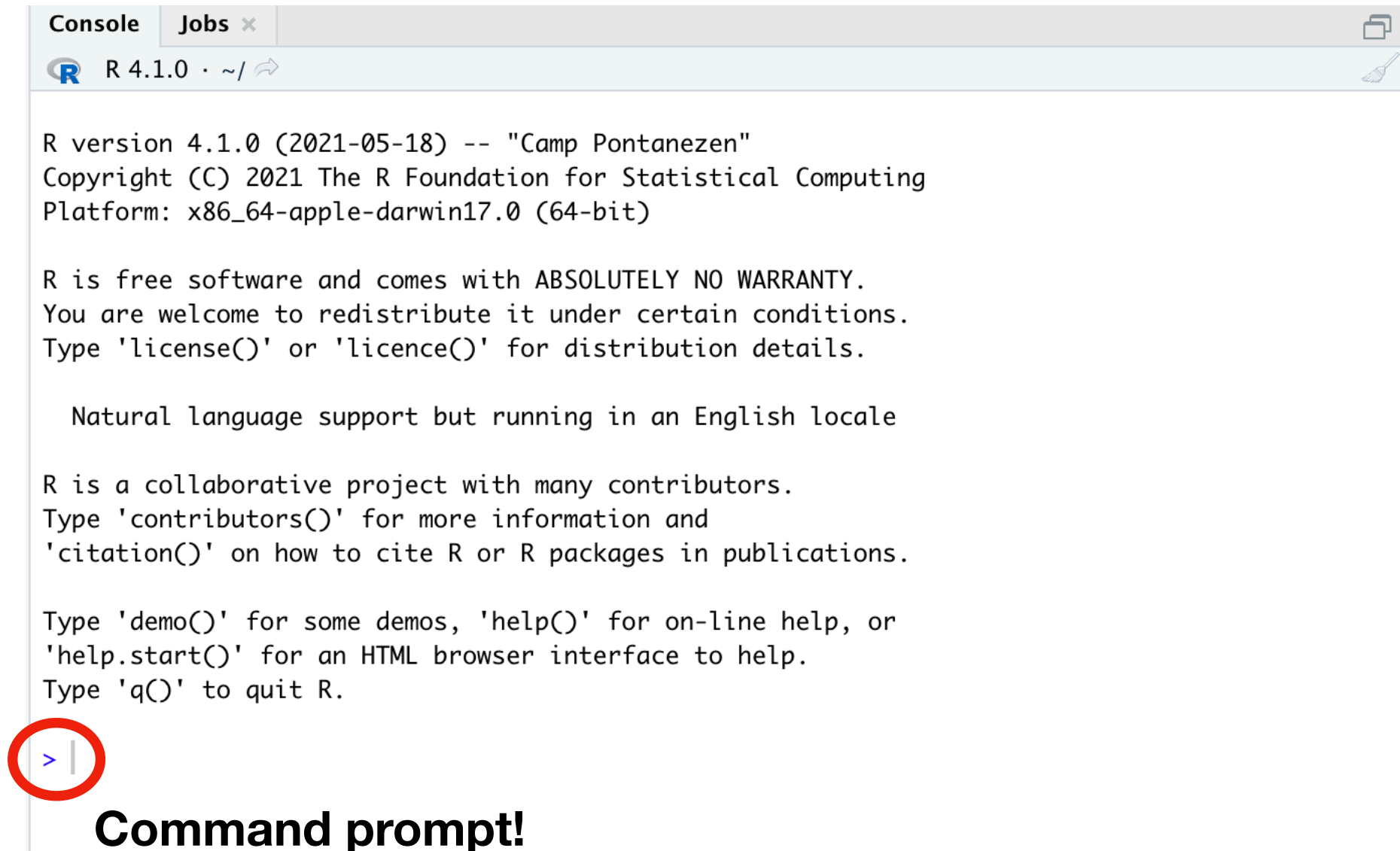
The Console window shows the R prompt and output, including the R license text and the message "Done with required packages!".

Adjusting Preferences

- Open RStudio's preferences:
 - Top bar > Tools > Global Options...
- We don't want your workspace saved!!!



Using Command Line



The screenshot shows the R console window with the title bar 'Console' and 'Jobs x'. The window displays the R version 4.1.0 startup message, including the copyright notice and platform information. The command prompt is highlighted with a red circle.

```
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

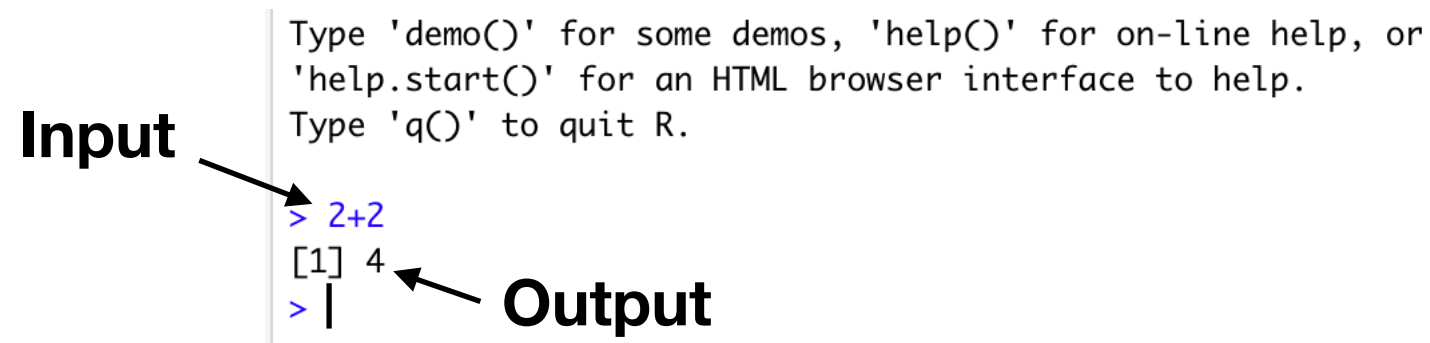
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```

Command prompt!

- Put your cursor here and enter something!



The diagram shows the R console with the command '2+2' entered at the prompt. The output '[1] 4' is displayed. Arrows point from the labels 'Input' and 'Output' to the respective parts of the console output.

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> 2+2
[1] 4
> |
```

Input → **Output**

***R* can function as a
basic calculator!**

R as a calculator

- R can act as a basic and scientific calculator

- Addition and Subtraction

```
> 2+2  
[1] 4  
> 2-2  
[1] 0
```

- Multiplication and Division

```
> 2*2      > 9/3  
[1] 4      [1] 3  
> 2*9      > 14/7+2  
[1] 18     [1] 4  
           > 14/(7+2)  
           [1] 1.555556
```

- Exponents and Logarithms

```
> 5^2      > log(234)  
[1] 25     [1] 5.455321  
> 6^-1     > log(243,base=3)  
[1] 0.1666667 [1] 5
```

It's a function!!



- e notation

```
> 1e-8  
[1] 1e-08  
> 8.39284e5  
[1] 839284  
> 837921597401782346  
[1] 8.379216e+17
```


Syntax in R

- Syntax is the set of rules that define what various combinations of symbols mean.
- Each computing language has a different set of rules of how you can arrange symbols to tell the computer what to do.

```
> 2+2
[1] 4
> 2 + 2
[1] 4
> 2      +      2
[1] 4
> 22+
+
+ 2
[1] 24
> 2,2,+
Error: unexpected ',' in "2,"
>
```

You can put spaces and tabs in between elements

But the elements need to be in the correct order!

And they must make sense to the computer (no commas allowed here!)

A lot of your errors, especially at first, will be syntax errors. This is ok, just fix it and try again!

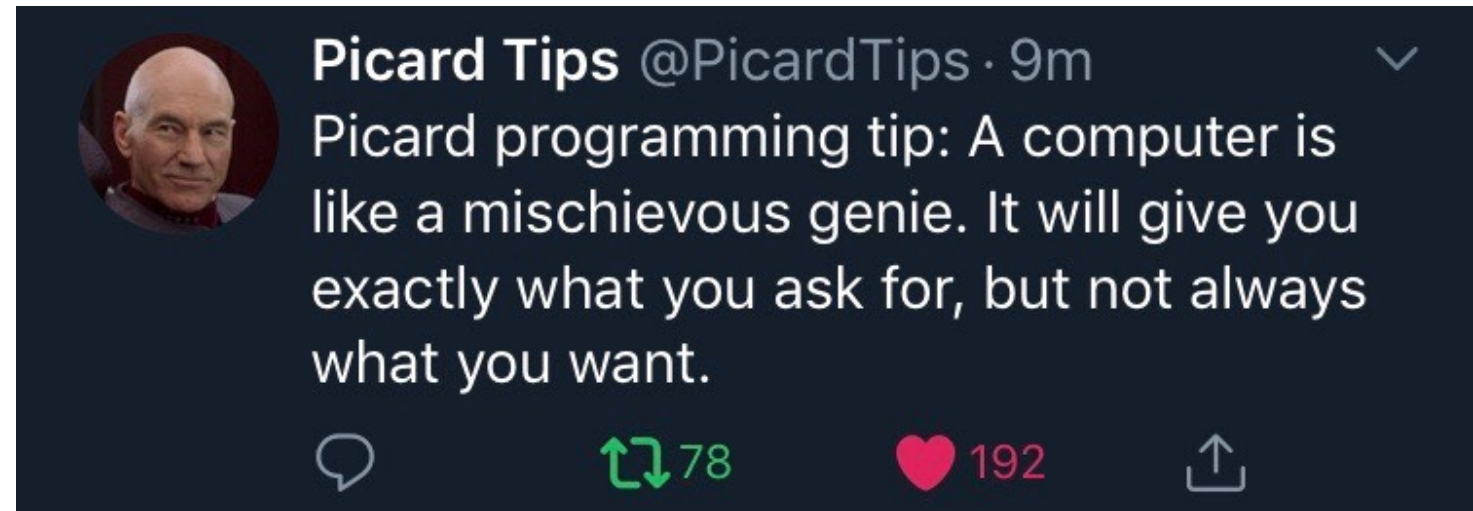
Let's just make R produce errors

- We're going to take a quick break to make R produce some errors. I make R errors all the time, constantly, even if it might look like I always know what I'm doing!
- Desensitize yourself to errors, they really aren't a big deal. You're not going to break anything.
- Feel free to swear at R, I don't care and neither does it.
- A huge part of learning to code is learning to tolerate being frustrated! It's ok and totally normal. Just don't let it consume you, take a break! Do something else! You'll get it eventually!

Some tips for using *R*

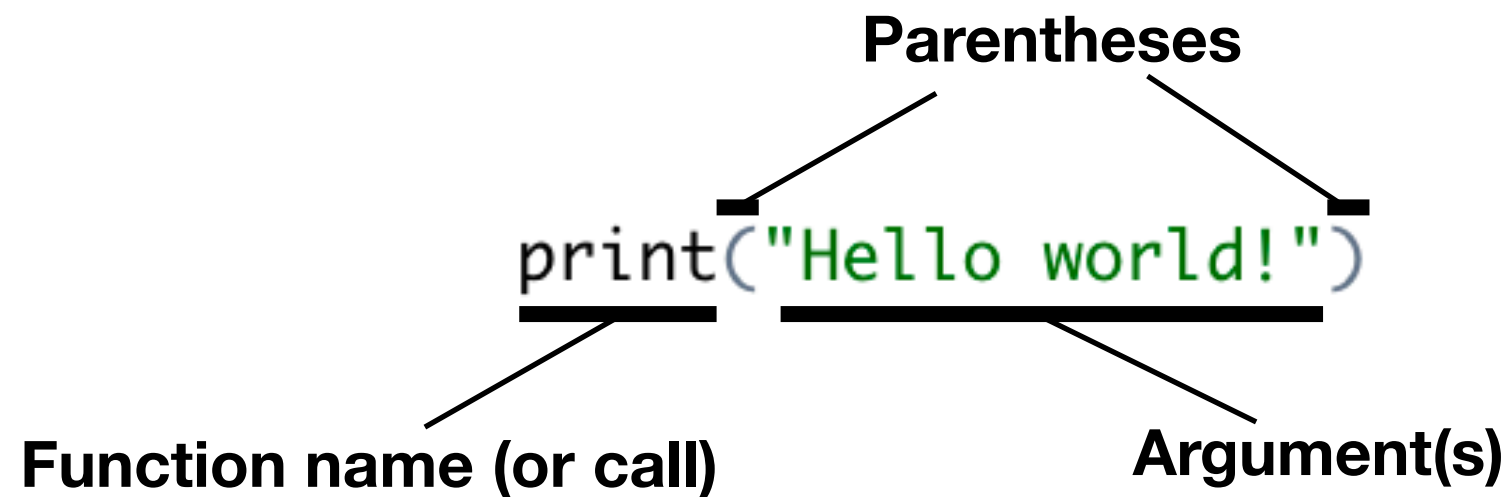
- Remember that computers do **EXACTLY** what you tell them to, but that's often not what you **THINK** you are telling them to do!
- Computers really only do what you tell them to do. It's always your fault. (It's always my fault too.)

A lot of your errors, especially at first, will be syntax errors. This is ok, just fix it and try again!



Functions in R

- Functions reference other bits of code that you can run by “calling” it (written by someone else or you)
- Syntax of a function in *R*:



- **Function name (or call):** the name of the function you desire to use, or how to “call” the function.
- **Parentheses:** Parentheses after the call is how R knows you want to call a function. You **MUST** put parentheses, even if the function takes no arguments, or else R will think you are trying to recall a variable!
- **Arguments:** options that you’d like to pass to the function.

Functions in R

- Use a function to create a vector:

```
> pop <- c(1, 9, 8, 2)
```

- Use another function to calculate the mean:

```
> mean(pop)
[1] 5
```

- You can nest functions within each other:

```
> mean(c(1, 9, 8, 2))
[1] 5
```

- Functions can have many arguments, separated by commas:

```
> blep <- seq(arg 1 = 1, arg 2 = 10, arg 3 = 0.5)
```

Creates a vector that is a sequence of numbers from 1 to 10 by 0.5

Environment	History	Connections	Tutorial
   Import Dataset ▾	 141 MiB ▾		
R ▾	Global Environment ▾		
Values			
a	2		
b	3		
pop	num [1:4] 1 9 8 2		

How many arguments a function can take is determined by the function!

R has lazy evaluation!

```
> blep <- seq(1, 10, 0.5)
```

When things go sideways... use help!

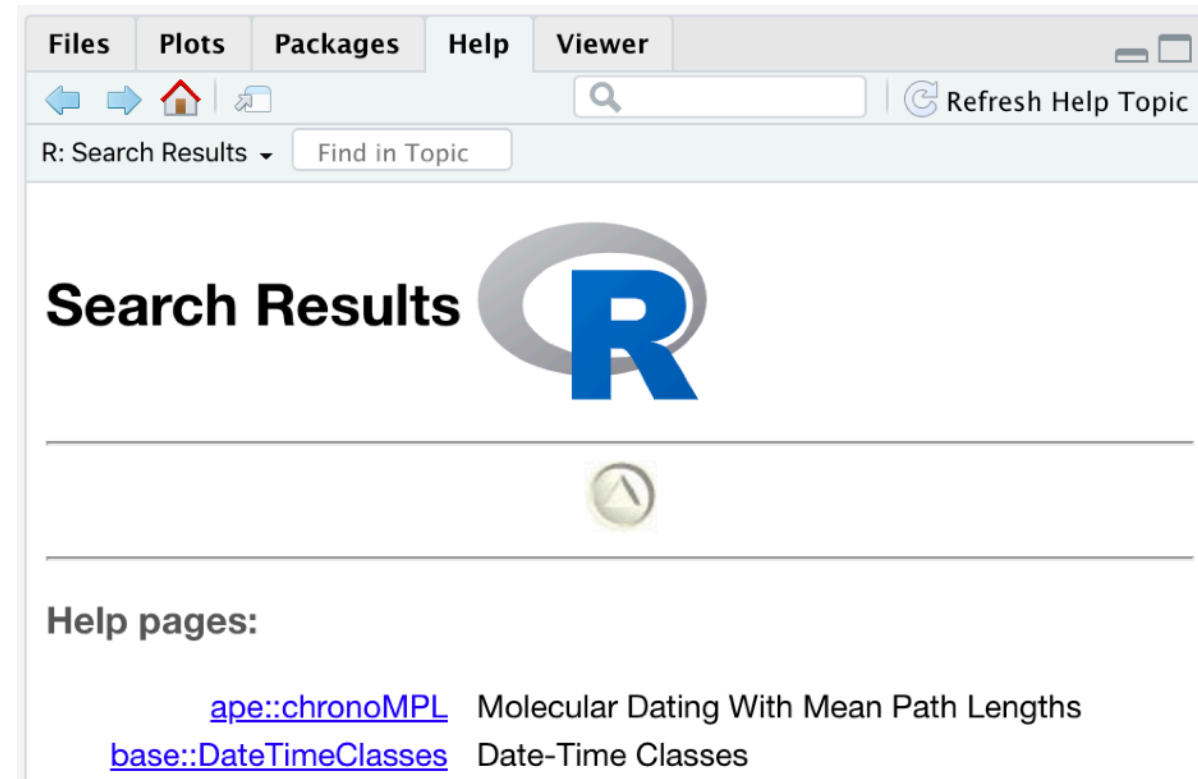
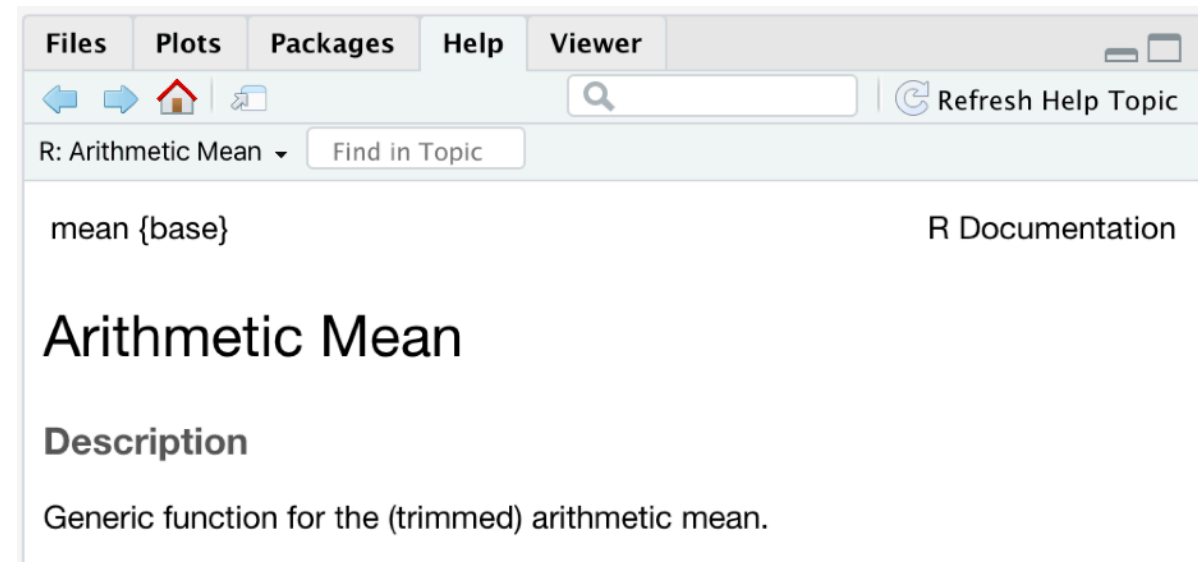
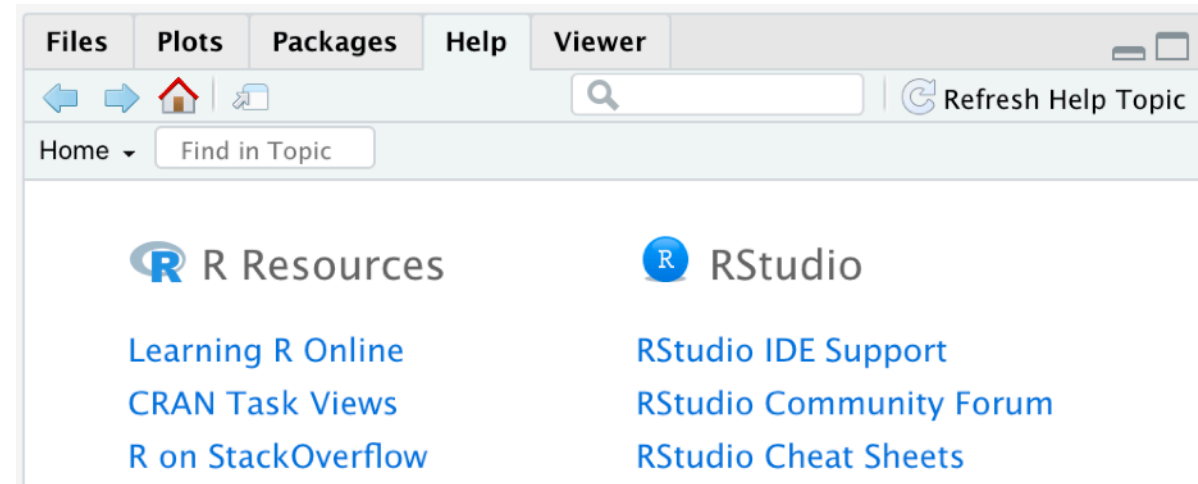
- Help documentation exists for each function in R.
- Help viewer is in the lower right-hand corner of RStudio as a tab.

- Access help documentation for a function:

```
> ?mean
```

- Do a keyword search:

```
> ??mean
```



Check Your Understanding

Have R calculate the square root of 164956 using the `sqrt()` function.
What is the correct output?

Correct answer

a) `[1] 406.1478`

b) `[1] 12.8841`

c) **Error: unexpected input in " $\sqrt{}$ "**

d) **Error in `sqrt(164956, 2)` : 2
arguments passed to 'sqrt' which
requires 1**

Error in c: I used the sign $\sqrt{164956}$, even though I understand what this means, R doesn't understand so returns an error.

Error in d: I put too many numbers (or arguments) in the `sqrt()` function so it's telling you you need one argument only!

Assigning variables and objects

- You'll notice that when you add $2+2$, R will give you an output, but it doesn't save or "store" that value anywhere. (Look at your environment, it is empty!)
- Assigning variables (objects): two methods

`> a <- 2`

- Assign arrow (less than sign and dash)

`> b = 3`

- Equal sign

**For historical reasons,
this is the preferred
method for variables...**

**...and this is the
preferred method
for arguments
inside functions.**

**Your environment tells you
what's in R's memory! This
is empty when R restarts!!**

- Assigned variables appear in your environment!

Environment	History	Connections	Tutorial
   Import Dataset ▾  129 MiB ▾ 			
R ▾	 Global Environment ▾		
Values			
a	2		
b	3		

Check Your Understanding

Syntax is important here! Using a space between the < and - will change the meaning of the command.

Try these: do they all have the same result? Why or why not?

a) `x<-2`

b) `x <- 2`

c) `x< - 2`

Does it matter which way the arrow points?

Try these: which work and which produce errors?

d) `7 -> j`

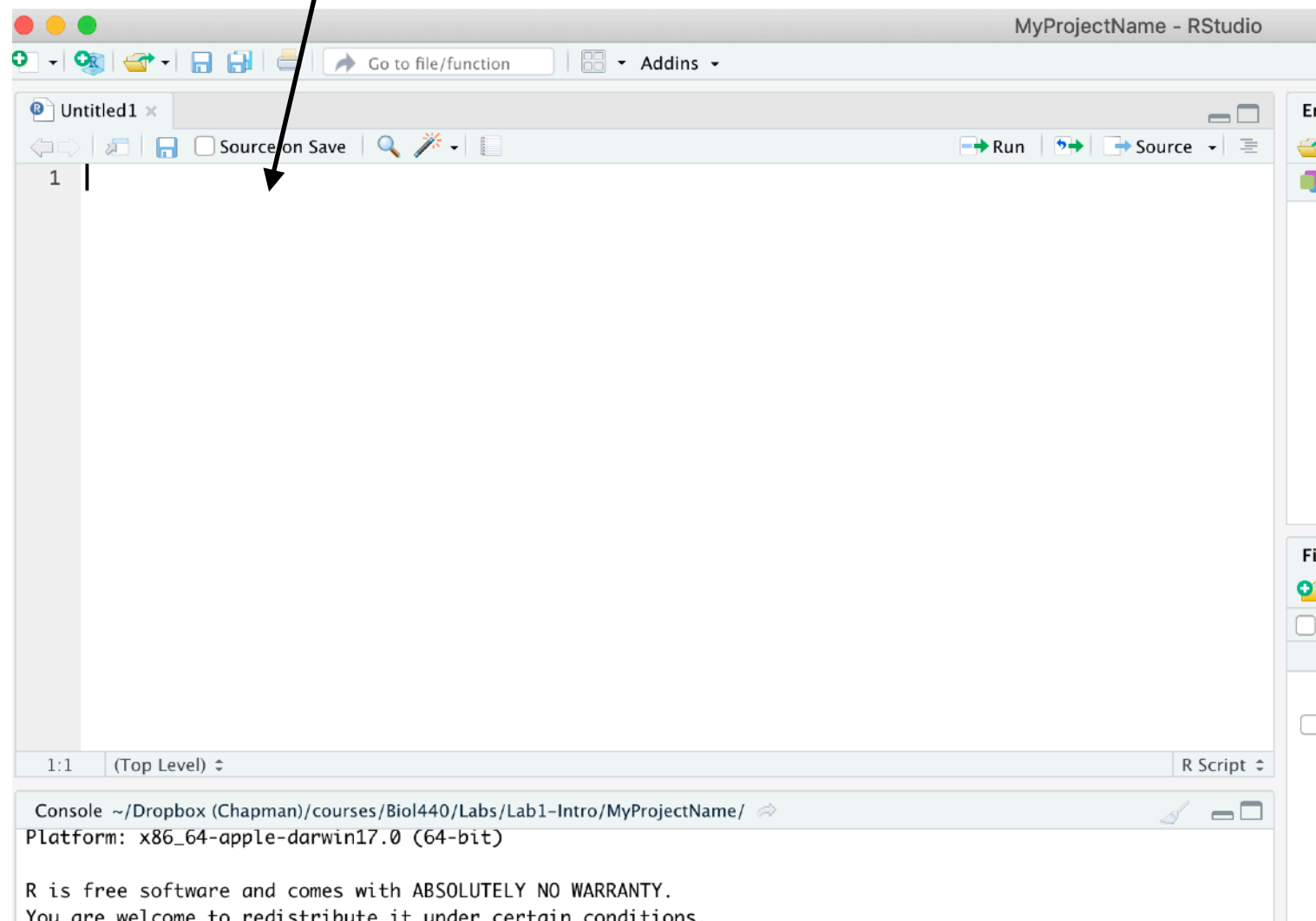
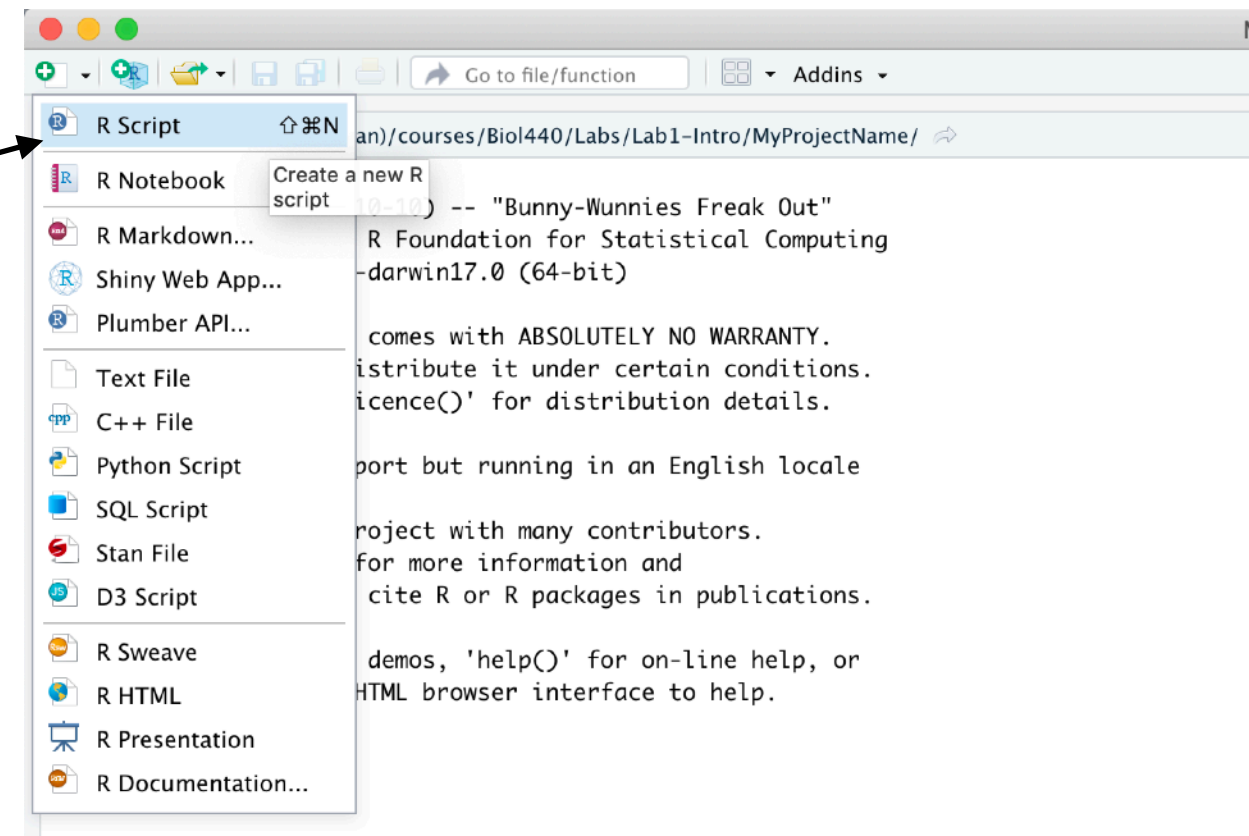
b) `7 <- j`

c) `j -> 7`

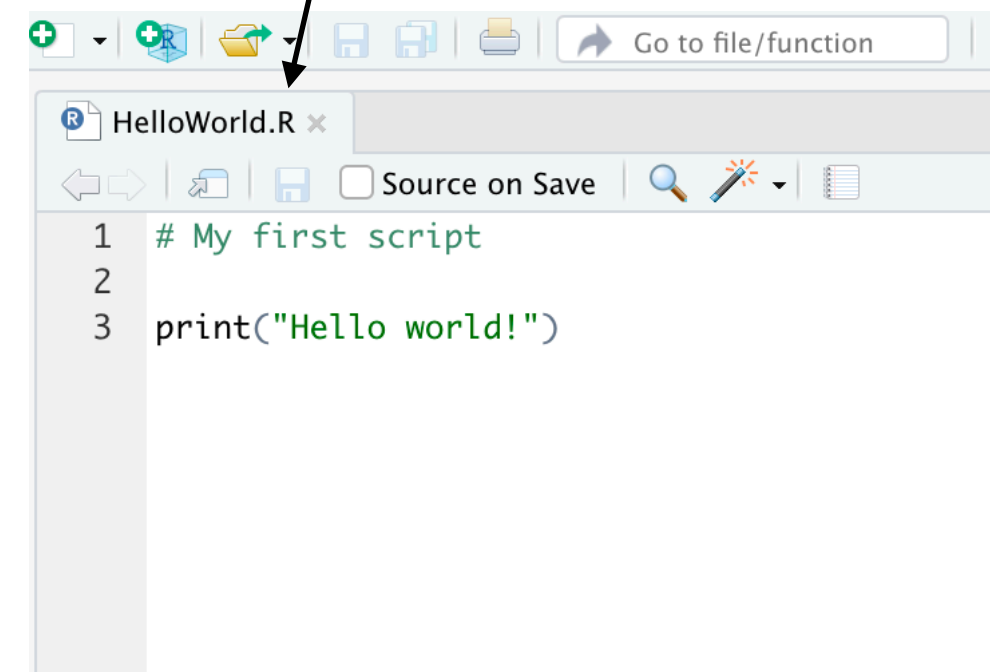
Your First R Script

Click on “New File” tab the in upper left and select “R Script”.

This will open a window in the script editor. You’re ready to code!



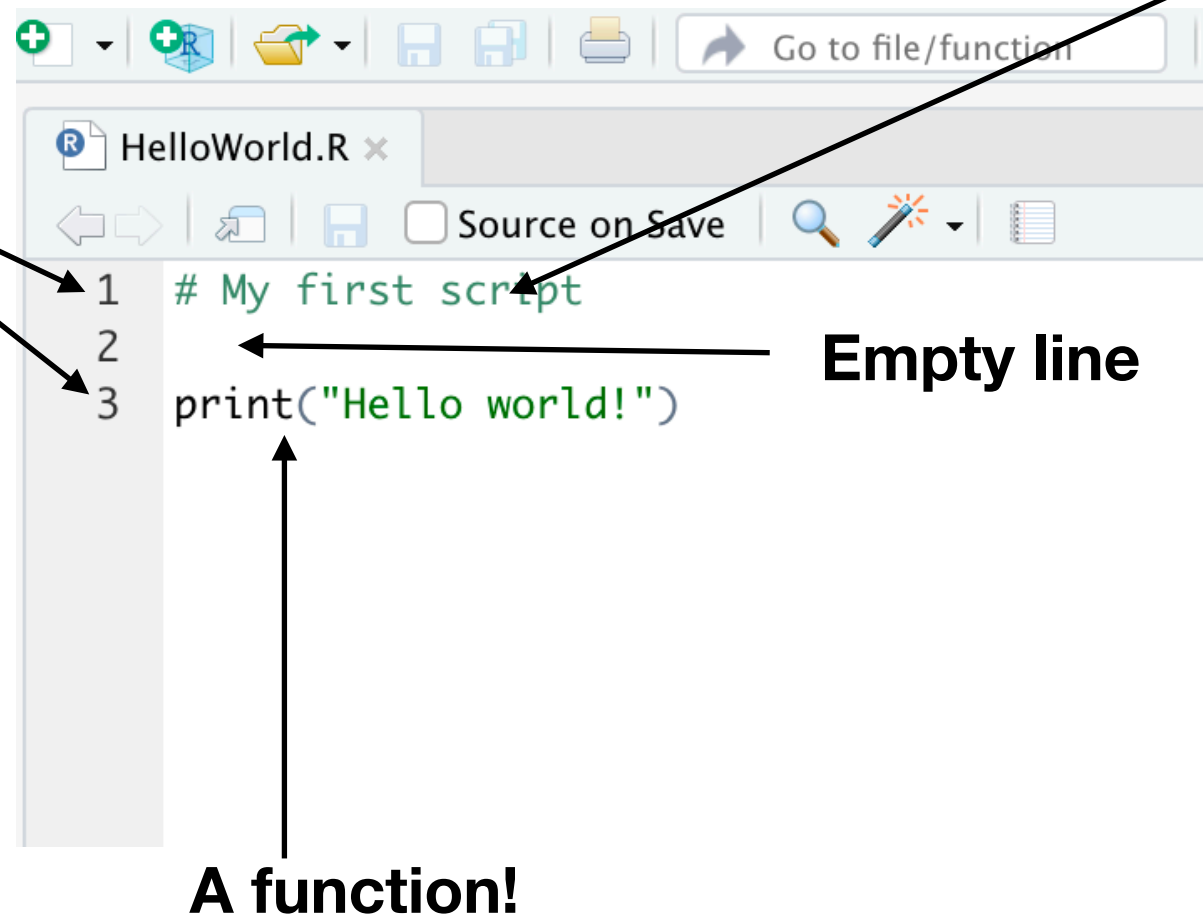
Be sure to save your script with a name that ends in .R



Your First R Script

Line numbers

- Useful for referencing specific bits of code
- Will show up in error messages

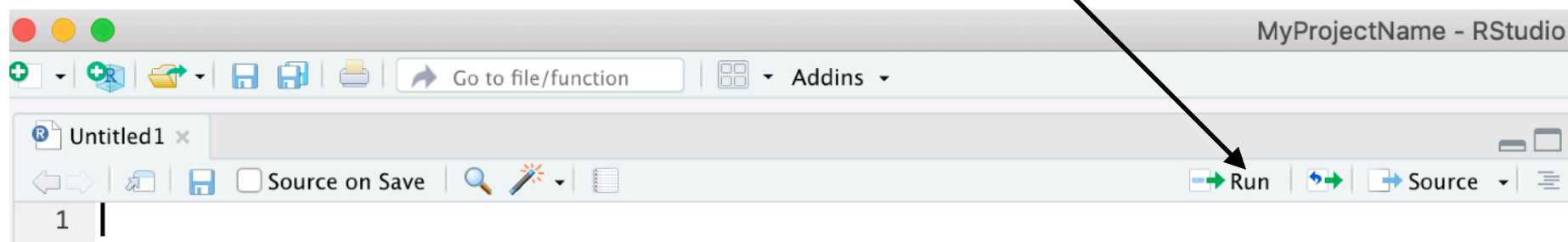


A "comment"

- Anything after a pound symbol #
- Skipped by R (not evaluated)
- Appears in green in script editor
- Really useful for documenting work

How to execute code in a script:

- Run each line by clicking the "Run" button



- Use a shortcut:

- Windows: Ctrl + Enter
- Mac: Command + Enter

Functions in R

Check your understanding!

What happens if you enter the numbers into mean without the **c()**? Or like this:

```
> mean(1, 9, 8, 2)
```

Why does it do this?



Functions in R

Check your understanding!

What happens if you enter the numbers into mean without the **c()**? Or like this:

```
> mean(1, 9, 8, 2)
```

Why does it do this?

Why would the mean not calculate correctly based on the help documentation for **mean()**?



More R Tips

- Use scripts to document code used during lecture! Make comments about what does and doesn't work, why errors occur, etc.
- Avoid using command line when possible, put it in a script!
- Make sure to write code that's evaluated from top to bottom without errors!
- You will probably get pretty frustrated. That's fine, just put it down and try again later.
- **Just because you get frustrated or produce errors from time to time doesn't mean you are bad at coding!**



Arun Karra
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me after the smallest inconvenience



Action Items

- 1. Complete Assignment 1.4**
- 2. Read Davies Chapter 4 for next time.**